

Diseases

Head smut and Rust are the major diseases occur in the Kodo Millets.

1) Head smut

Symptoms : Head smut is seed borne disease and infected inflorescence is deformed and produces gall-like swelling on the stem, the nodes of young shoots and in the axils of the older leaves infected panicle may older leaves Infected panicle may be enclosed in the flag leaf and may not emerge fully.

Management: The pathogen being seed borne can be checked by seed treatment with Thiram or Ceresan @ 2.5 g/kg of seed and soaking seeds in hot water at 55 degree for 7-12 minutes. The variety GPUK 3 is tolerant to head smut.

2) Rust

Symptoms: Brown pustules are seen on leaves. Hinders the photosynthesis and cause considerable loss in yield.

Management: Spray 0.2% solution of Mancozeb 75 WP

Pests: Shoot Fly is the major pest occurs in the Kodo Millet.

Shoot fly Symptom

Feeding the central shoot starts drying and shows the typical appearance of 'dead heart' in early stage and profuse tillering in later stage.

Management: Early sowing of crop with the onset of monsoon is beneficial. Sow the crop before 2nd fortnight of July. Another important

practice is to increase the seed rate (1.5 times the recommended seed rate). Apply Phorate 10G @ 15kg/ha in the soil at time of field preparation or Carbofuran 3G granules @ 30 kg/ha in furrows or as broadcast before sowing. Need-based application of insecticides.

Harvesting: Normally crop is ready for harvest in 100-120 days after sowing, depends upon variety and local weather condition.

Yield: Grain Yield 15-18 q/ha and
Straw 30-40 q/ha.

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Kodo Millet



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Kodo Millet (*Paspalum scrobiculatum* L.) is the native to the tropical and subtropical regions of South America and domesticated in India 3,000 years ago. Kodo Millet is extensively grown on the poorest of soils throughout India but is probably not cultivated to any extent elsewhere. It is reputed to be extremely hardy, drought resistant and grows on stony or gravelly soils, needs little rainfall which would not support other crops. It is relatively long duration crop require 100-140 days to mature compared with 65-120 days for other millets. Short duration varieties are being developed. Kodo Millet is an annual tufted grass that grows up to 90 cm high. The grain is enclosed in hard, corneous, persistent husks that are difficult to remove. The grain may vary in colour from light red to dark grey. Kodo Millet comes up in poor soils, and responses well to improved management practices. Its cultivation is prominent in Madhya Pradesh, Karnataka, Tamil Nadu, Telangana, Andhra Pradesh, Orissa, Bihar and Maharashtra. Kodo millet contains good amounts of Protein, carbohydrate, fat, dietary fiber, and minerals. The grain is recommended as a substitute for rice, for patients who are suffering from diabetes. In spite of having significant nutrients, it is cultivated in less fertile land. But, as a result of successful efforts of the ICAR-IIMR and AICRP of small millets in crop improvement, today Kodo Millet production has improved greatly.

Local Names of Kodo Millet

Hindi	Kodon, Kodo
Telugu	Arika, Arikelu
Kannada	Haraka
Marathi	Kodra, Kodru
Malayalam	varaku
Tamil	Varaku
Bengali	Kodo
Gujarathi	Kodra

State wise Recommended Cultivars of Kodo-millet

State	Cultivars of Kodo Millet
Madhya Pradesh	JK 137, DSP-9-1, JK 98, JK 106, JK 439, RBK 155, JK 13, JK 65, JK 48, RK 390-25, GPUK 3, TNAU-86
Tamil Nadu	KMV 20 (Bamban), CO 3, TNAU 86, GPUK 3, RK 390-25, JK 13
Gujarat	GK 1, GK 2, GPUK 3, RK 390-25, JK 13, JK 65
Chhattisgarh	RBK 155 and JK 439, Indira Kodo-1, JK -48, GPUK 3, JK-65, JK 98, RK 390-25, TNAU-86
Karnataka	GPUK 3, RBK 155, RK 390-25, TNAU-86

Soil & Climate: Kodo Millet can be grown in gravelly and stony upland poor soil to loam soil. Deep, loamy, fertile soil, rich in organic matter results in better yield. Kodo Millet is drought tolerant crop can be grown in dry and warm climate with rainfall of 500-600mm.

Time of Sowing:

Kharif-Middle of June to last week of July,

Rabi-September to October,

Sowing with the onset of monsoon is beneficial

Spacing: Distance between row to row is 22.5 cm and plant to plant distance should be 10 cm. Seed should be sown at 3- 4 cm in depth.

Seed rate: 8-10 kg/ha for line sowing

15 kg/ha for broadcasting

Manures and fertilizers:

Apply Compost or farmyard manure @ 5-7.5 tonnes/ha about a month before sowing. Generally fertilizer recommended to get a good crop are 40 kg Nitrogen, 20 kg P₂O₅ and 20 kg K₂O per ha in all states except Tamil-Nadu, Madhya Pradesh and Chhattisgarh where 40 kg N and 20 kg P₂O₅ per ha. Soil test based fertilizers application is recommended. Apply entire quantity of P₂O₅ and half of Nitrogen at the time of sowing and remaining half of Nitrogen at 35-40 days after sowing.

Weeding and Intercultural Operations: Two inter cultivations and one hand weeding in line sown crop is recommended. Intercultural operation using a tyne-h arrow when crop is 30 days old is recommended. In broadcast crop 1st weeding after 15 - 20 days after emergence of seedling and 2nd weeding 15-20 days after 1st weeding is recommended for better yield.

Irrigation: Kodo Millet requires minimum irrigation; First irrigation can be given at 25-30 days after sowing and second at 40-45 days after sowing. If dry spell prevails for longer period, irrigations are required every 4-7 days depending on the severity of the drought and type of soil.

Cropping System

a) Inter cropping: Inter cropping with legumes like Pigeon Pea, Green gram, Black Gram & Soyabean in 2:1 proportion found profitable in many states.

b) Crop Rotation: Rotation with crops like Soyabean or Niger in Madhya Pradesh found to be sustainable.